

AP CALCULUS AB – UNIT 4

Contextual Applications of Differentiation

AP-Style Review and Exam Practice



Name: _____ Class: _____ Date: _____

Directions

Section I Part A contains 16 non-calculator multiple-choice questions. Section I Part B contains 8 calculator-permitted multiple-choice questions. Section II contains four free-response questions. Show setup, exact reasoning, units, and justification.

Section I – Part A: No Calculator

1. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

- (A) months per dollar (B) dollars per month
(C) dollars (D) dollars per month squared

2. A particle has position $s(t) = 3t^3 - 4t^2 + 3t$. Find its acceleration $a(t)$.

- (A) $9t^2 - 8t + 3$ (B) 18
(C) $t(3t^2 - 4t + 3)$ (D) $18t - 8$

3. A particle's velocity is $v(t) = (t - 1)(t - 2)$. On which interval does it move left?

- (A) $(-\infty, 1) \cup (2, \infty)$ (B) $(1, 2)$
(C) $(-\infty, 1)$ (D) $(2, \infty)$

4. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

- (A) $Q'(a) > 0$ and $Q''(a) > 0$ (B) $Q'(a) > 0$ and $Q''(a) < 0$
(C) $Q'(a) < 0$ and $Q''(a) < 0$ (D) $Q'(a) < 0$ and $Q''(a) > 0$

5. A circle's radius increases at 4 cm/s. How fast is its area increasing when $r = 4$ cm?

- (A) 32π cm/s (B) 32π cm²/s
(C) 16π cm²/s (D) 33π cm²/s

6. A 10-m ladder has base $x = 8$ m moving away at 2 m/s. Find the rate of the top.

- (A) 2 m/s (B) 2.667 m/s
(C) -2 m/s (D) -2.667 m/s

7. Given $f(3) = 4$ and $f'(3) = 1$, use linearization to approximate $f(\frac{16}{5})$.

- (A) 5 (B) $\frac{19}{5}$
(C) $\frac{1}{5}$ (D) $\frac{21}{5}$

8. Use $f(x) = \sqrt{x}$ linearized at $x = 16$ to approximate $\sqrt{16 + 0.01}$.

- (A) $\frac{3201}{800}$ (B) $\frac{401}{100}$
(C) $\frac{33}{8}$ (D) $\frac{3199}{800}$

9. Before applying L'Hopital's Rule to a quotient limit, which must be verified?

- (A) The quotient is continuous at the target (B) The numerator derivative is zero
(C) The numerator and denominator are polynomials (D) The quotient has form $0/0$ or ∞/∞

10. Evaluate $\lim_{x \rightarrow 0} \frac{e^{4x} - 1}{x}$.

- (A) e^4 (B) 4
(C) 1 (D) 0

11. Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 1}{e^x}$.

- (A) 4 (B) $+\infty$
(C) The limit does not exist (D) 0

12. If $v(t) > 0$ and $a(t) < 0$, which statement is true?

- (A) The particle moves right and speeds up (B) The particle moves left and speeds up
(C) The particle moves left and slows down (D) The particle moves right and slows down

13. If $C(t)$ is measured in dollars and t in months, what are the units of $C'(6)$?

- (A) dollars per month squared (B) dollars
(C) months per dollar (D) dollars per month

14. A particle has position $s(t) = 4t^3 - t^2 + t$. Find its acceleration $a(t)$.

- (A) 24 (B) $t(4t^2 - t + 1)$
(C) $24t - 2$ (D) $12t^2 - 2t + 1$

15. A particle's velocity is $v(t) = (t - 2)(t - 3)$. On which interval does it move left?

- (A) $(-\infty, 2) \cup (3, \infty)$ (B) $(3, \infty)$
(C) $(-\infty, 2)$ (D) $(2, 3)$

16. A quantity Q is increasing at a decreasing rate at $t = a$. Which signs are correct?

- (A) $Q'(a) > 0$ and $Q''(a) < 0$ (B) $Q'(a) > 0$ and $Q''(a) > 0$
(C) $Q'(a) < 0$ and $Q''(a) < 0$ (D) $Q'(a) < 0$ and $Q''(a) > 0$

Section I – Part B: Calculator Permitted

17. A particle has $s(t) = t^3 - 2t^2 + t$. Use a calculator to find $v(1.64)$.

- (A) 0.672 (B) 3.509
(C) 5.84 (D) 2.509

18. Using $f(4.03) \approx 1.3938$ and $f'(4.03) \approx 0.2481$, approximate $\ln(3.978)$.

- (A) 1.3418 (B) 1.6419
(C) 1.38 (D) 1.3809

19. Use a calculator to estimate $\frac{e^{0.88} - 1}{0.88}$. Which value is closest?

- (A) 2.411 (B) 0.88
(C) 1.603 (D) 2.603

20. A sphere has radius $r = 5.1$ cm and $dr/dt = -0.4$ cm/s. Find dV/dt numerically.

- (A) $-25.635 \text{ cm}^3/\text{s}$ (B) $-130.741 \text{ cm}^3/\text{s}$
(C) $-129.741 \text{ cm}^3/\text{s}$ (D) $-222.259 \text{ cm}^3/\text{s}$

21. At $t = 2.6$, a particle has velocity 5.0 m/s and acceleration -3.6 m/s^2 . Which best describes its motion?

- (A) At rest (B) Moving right and slowing down
(C) Moving left and slowing down (D) Moving right and speeding up

22. A particle has $s(t) = t^3 - 2t^2 + t$. Use a calculator to find $v(2.12)$.

- (A) 6.003 (B) 7.003
(C) 8.72 (D) 2.659

23. Using $f(3.1) \approx 1.1314$ and $f'(3.1) \approx 0.3226$, approximate $\ln(3.101)$.

- (A) 1.1324 (B) 1.1317
(C) 1.454 (D) 1.13

24. Use a calculator to estimate $\frac{e^{0.34} - 1}{0.34}$. Which value is closest?

- (A) 2.191 (B) 0.34
(C) 1.191 (D) 1.405

Section II – Free Response

Questions 1–2: Calculator permitted. Questions 3–4: No calculator.

FRQ1 – Straight-line motion

A particle has velocity $v(t) = t^2 - 4t + 3$ for $0 \leq t \leq 5$.

(a) Find the times when the particle changes direction. [3 points]

(b) Determine intervals on which the particle moves right. [2 points]

(c) Find acceleration and determine when speed increases. [3 points]

FRQ2 – Related rates: conical tank

A cone has radius-to-height ratio $r/h = 1/3$. Water enters at $2 \text{ m}^3/\text{min}$.

(a) Write volume as a function of height only. [2 points]

(b) Find dh/dt when $h = 6$. [3 points]

(c) Interpret the sign and units. [1 points]

FRQ3 – Linear approximation

Let $f(x) = \sqrt{x}$.

(a) Find the linearization at $x = 16$. [2 points]

(b) Use it to approximate $\sqrt{16.3}$. [2 points]

(c) Determine whether the approximation is an overestimate or underestimate and justify using concavity. [2 points]

FRQ4 – L'Hopital and interpretation

Consider $\lim_{x \rightarrow 0} \frac{\ln(1 + 3x)}{x}$.

(a) Verify the indeterminate form. [1 points]

(b) Apply L'Hopital's Rule to evaluate the limit. [2 points]

(c) Interpret the limit as a derivative. [2 points]